Groww AI Discovery Engine

⚙️ Pipeline Breakdown

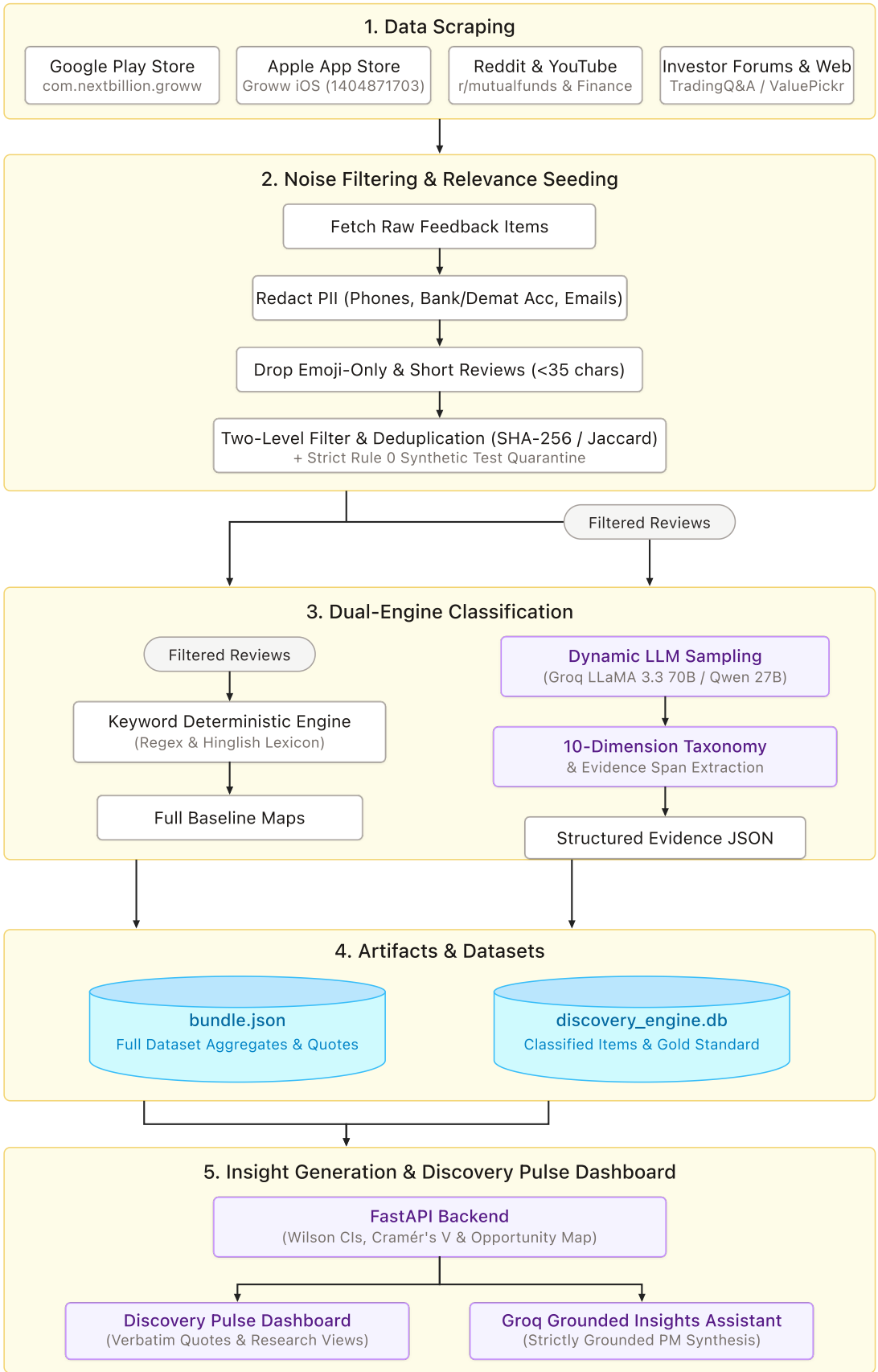
- **Data Scraping:** The pipeline begins by fetching real user feedback across multi-channel public platforms—Google Play Store (com.nextbillion.groww), Apple App Store (Groww: Stocks & Mutual Funds), targeted Reddit communities (r/mutualfunds, r/personalfinanceindia, r/IndiaInvestments, r/IndianStreetBets, r/groww), YouTube video comments (official Groww mutual fund explainers and retail finance channels), investor help forums (TradingQ&A by Zerodha, ValuePickr), and financial web queries using Python collectors, Apify actors, and Playwright scraper scripts.

- **Noise Filtering & Relevance Seeding:** Before analysis, raw text is aggressively filtered. PII (phone numbers, emails, bank/demat account numbers, PAN identifiers) is redacted, low-quality reviews (emoji-only, <35 characters, alpha-character density <40%) are dropped, hash-based deduplication (exact SHA-256 and Jaccard token similarity ≥ 0.85) removes duplicates, and a **Two-Level Relevance Filter** isolates retention-relevant signals: Level 1 filters out equity trading and crypto noise, while Level 2 isolates active SIP drop, pause, failure, or regret signals (*"stop sip"*, *"failed debit"*, *"bounce charge"*, *"negative returns"*, *"xirr wrong"*, *"how to pause"*), alongside strict Rule 0 synthetic test quarantine.

- **Dual-Engine Classification:**
    - **Keyword Deterministic Engine:** Fast, regex-based matching with Hinglish and Hindi vocabulary support that buckets all incoming reviews into baseline friction themes, lifecycle stages, and drop types.
    - **Dynamic LLM Engine:** On-demand Groq LPU API (powered by LLaMA 3.3 70B and Qwen 27B) that classifies reviews into a structured **10-Dimension Research Taxonomy** (*drop_type*, *sip_status*, *trigger*, *stage_of_mention*, *in_app_confusion_object*, *mandate_subcluster*, *stated_expectation*, *outside_app_action*, *disconfirming_pass_attribution*, *segment_cue*) with verbatim evidence extraction and confidence scores.

- **Artifacts & Datasets:** The classified dataset and aggregated metrics are persisted in SQLite (discovery_engine.db) across normalized tables (raw_items, tagged_items), as well as static JSON and CSV artifacts (bundle.json, opportunity_areas.csv, and unsure_question_bank_v2.json), enabling a stateless, decoupled backend architecture and high-performance serverless deployments.

- **Insight Generation & UI:** The FastAPI API server reads active artifacts, calculates Wilson 95% confidence intervals, Cramér's V channel independence scores, and multi-factor opportunity rankings to construct the **Ranked Opportunity Map (10 Defensible Opportunity Spaces)** and **105 Audited Golden Questions**. The **Discovery Pulse Dashboard** surfaces verbatim user quotes and powers an interactive **Groq AI Insights Assistant** strictly grounded on these artifacts without hallucinations.


1. Data Scraping
Google Play Store
com.nextbillion.groww
Apple App Store
Groww iOS (1404871703)
Reddit & YouTube
r/mutualfunds & Finance
Investor Forums & Web
TradingQ&A / ValuePickr
2. Noise Filtering & Relevance Seeding
Fetch Raw Feedback Items
Redact PII (Phones, Bank/Demat Acc, Emails)
Drop Emoji-Only & Short Reviews (<35 chars)
Two-Level Filter & Deduplication (SHA-256 / Jaccard)
+ Strict Rule 0 Synthetic Test Quarantine
Filtered Reviews
3. Dual-Engine Classification
Filtered Reviews
Keyword Deterministic Engine
(Regex & Hinglish Lexicon)
Full Baseline Maps
Dynamic LLM Sampling
(Groq LLaMA 3.3 70B / Qwen 27B)
10-Dimension Taxonomy
& Evidence Span Extraction
Structured Evidence JSON
4. Artifacts & Datasets
bundle.json
Full Dataset Aggregates & Quotes
discovery_engine.db
Classified Items & Gold Standard
5. Insight Generation & Discovery Pulse Dashboard
FastAPI Backend
(Wilson CIs, Cramér's V & Opportunity Map)
Discovery Pulse Dashboard
(Verbatim Quotes & Research Views)
Groq Grounded Insights Assistant
(Strictly Grounded PM Synthesis)